A REPORT ON

DELOITTE AUSTRALIA DATA ANALYTICS JOB SIMULATION

(Completed via Forage Virtual Job Simulation Platform)

Submitted in partial fulfilment of the requirements for

B.Tech – Computer Science and Engineering

Submitted by

Garima Singh

Roll No.: 25SCS1003003257

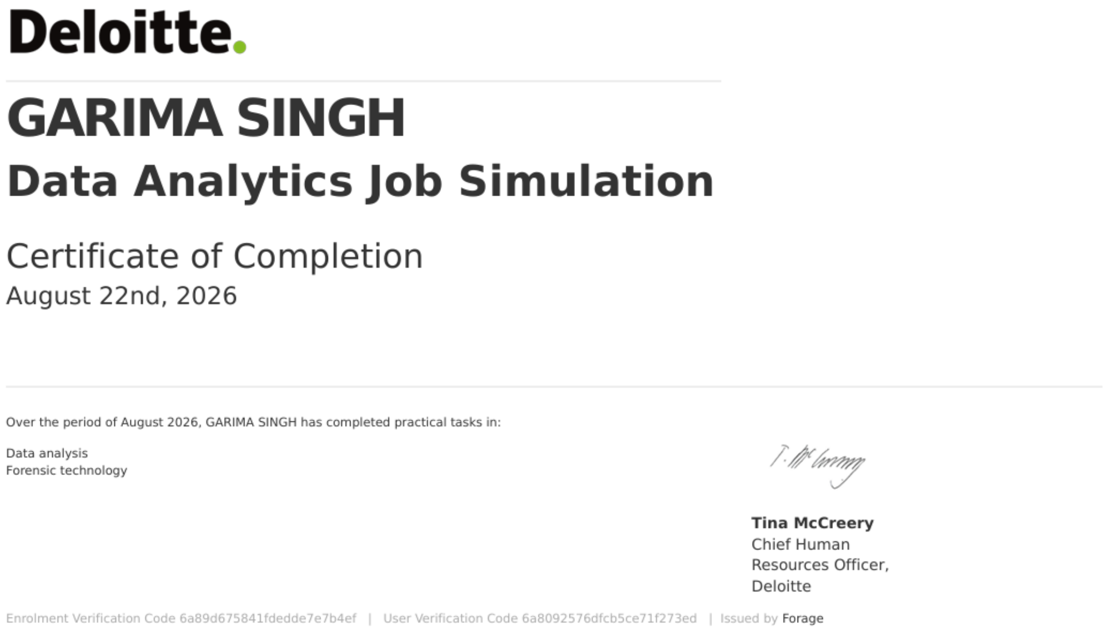
Session: 2025–29

IILM UNIVERSITY, GREATER NOIDA

Department of Computer Science and Engineering

Internship Certificate

The certificate of completion issued by Deloitte Australia through the Forage platform for the Data Analytics Job Simulation is provided below.


Deloitte.

GARIMA SINGH

Data Analytics Job Simulation

Certificate of Completion

August 22nd, 2026

Over the period of August 2026, GARIMA SINGH has completed practical tasks in:

Data analysis
Forensic technology

Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code 6a89d675841fdedde7e7b4ef | User Verification Code 6a8092576dfcb5ce71f273ed | Issued by Forage

Declaration

I, Garima Singh, a student of B.Tech – Computer Science and Engineering (Session 2025–29), Roll No. 25SCS1003003257, at IILM University, Greater Noida, hereby declare that the work presented in this report titled "Deloitte Australia Data Analytics Job Simulation" is a genuine record of the tasks I completed through the Forage virtual job simulation platform.

This report has been prepared solely on the basis of my own work carried out during the simulation, and the content presented herein has not been copied from any other source except where explicitly referenced. I understand that this was a virtual job simulation and not a period of direct employment with Deloitte.

Place: Greater Noida
Date: [To be filled]

Garima Singh

Acknowledgement

I would like to express my sincere gratitude to IILM University, Greater Noida, for encouraging students to undertake industry-oriented learning experiences such as virtual job simulations, which provide valuable exposure to real-world professional tasks.

I am thankful to the Department of Computer Science and Engineering for their continuous support and guidance throughout my academic journey, which enabled me to pursue and complete this simulation successfully.

I also extend my appreciation to Deloitte Australia and the Forage platform for designing and providing the Data Analytics Job Simulation, which allowed me to apply analytical tools such as Tableau and Microsoft Excel to practical, industry-relevant problems and gain a better understanding of the data analytics profession.

Finally, I would like to thank my family and peers for their constant encouragement during the completion of this simulation and the preparation of this report.

Garima Singh

Table of Contents

Introduction

This report presents a detailed account of my learning experience during the Deloitte Australia Data Analytics Job Simulation, a virtual job simulation completed through the Forage platform. The simulation was designed to provide students with exposure to the kind of analytical tasks performed by data analysts at Deloitte, using industry-standard tools such as Tableau and Microsoft Excel.

The simulation was undertaken as part of my effort to gain practical, industry-relevant skills alongside my academic curriculum in B.Tech – Computer Science and Engineering. It is important to clarify that this was a self-paced virtual simulation and not a period of direct employment with Deloitte; no supervisor was assigned, and the tasks were completed independently based on instructions provided on the Forage platform.

This report documents the two tasks I completed as part of the simulation — a data analysis task involving telemetry data visualisation in Tableau, and a forensic technology task involving the classification of equality scores in Microsoft Excel — along with the tools used, the skills developed, and the learning outcomes achieved.

About Deloitte

Deloitte is one of the largest professional services networks in the world, providing services in the areas of audit and assurance, consulting, financial advisory, risk advisory, tax, and legal services to clients across a wide range of industries. Deloitte Australia is the Australian member firm of the Deloitte global network and offers similar services to clients within Australia and the surrounding region, including work in the field of data analytics and forensic technology.

Within its consulting and forensic practices, Deloitte works with data analytics teams that help organisations interpret operational data, identify patterns, and support decision-making. The Data Analytics Job Simulation was created to give students a representative sense of the type of analytical work carried out in such teams, using simplified but realistic datasets and business scenarios.

About Forage

Forage is an online platform that partners with global companies, including Deloitte, to offer free virtual job simulations to students. These simulations are designed to replicate the type of tasks that employees in a given role or department would typically perform, allowing students to gain practical exposure to a company's work culture and technical expectations without being formally employed.

Each simulation on Forage is self-paced and includes task instructions, background reading material, and practical exercises, followed by feedback once submitted. On successful completion of all tasks in a simulation, Forage issues a certificate of completion, which serves as evidence of the skills practised during the simulation.

The Deloitte Australia Data Analytics Job Simulation that I completed was one such program, focused specifically on tasks relevant to data analysis and forensic technology roles within Deloitte's consulting practice.

Internship Overview

The Deloitte Australia Data Analytics Job Simulation was completed entirely online through the Forage platform, in a self-paced virtual format. The simulation consisted of two independent tasks, each focused on a different aspect of data analytics work: a data visualisation task using Tableau, and a data classification task using Microsoft Excel.

Particular	Detail
Organisation	Deloitte Australia
Platform	Forage
Program Name	Data Analytics Job Simulation
Mode	Virtual Job Simulation (self-paced, online)
Domain	Data Analytics
Duration	[To be filled]
Supervisor	Not applicable — self-paced virtual simulation with no assigned supervisor
Certificate Date	August 22, 2026

As indicated above, the simulation did not involve an assigned supervisor or a fixed internship duration in the traditional sense, since it was a self-directed virtual program. The certificate of completion, shown on Page 2 of this report, records the date of completion as August 22, 2026.

Objectives of the Internship

The Data Analytics Job Simulation was undertaken with the following objectives in mind:

- To gain hands-on experience with industry-standard data analytics tools, specifically Tableau and Microsoft Excel.
- To understand how raw operational data, such as equipment telemetry readings, can be transformed into meaningful visual insights.
- To learn how conditional logic and formulas can be applied to classify data for forensic or compliance-related analysis.
- To develop familiarity with the type of analytical tasks performed by data analysts within a professional consulting environment.
- To practise documenting and submitting analytical work in a structured and professional manner, including version control and submission through GitHub.

Tools and Technologies Used

The following tools were used to complete the two tasks in this simulation. No programming languages, database query tools, or machine learning frameworks were used, as the tasks were designed around visual analytics and spreadsheet-based data classification.

Tool	Purpose
Tableau	Creating calculated measures, bar charts, and an interactive dashboard for the telemetry data analysis task.
Microsoft Excel	Classifying equality scores using a conditional formula in the forensic technology task.
GitHub	Hosting a public repository for submission and documentation of internship-related work.

Task 1 – Data Analysis

The first task in the simulation required me to work with Tableau to analyse telemetry data collected from factory equipment belonging to a fictional client company, Daikibo. The objective of this task was to identify how much potential downtime each factory and device type had experienced, based on recorded machine health statuses, and to present these findings through an interactive dashboard.

The specific instructions for this task were to import the dataset into Tableau, create a calculated field to quantify downtime, build two bar charts, combine them into a dashboard with filtering functionality, and identify the factory with the highest downtime.

Daikibo Telemetry Dataset

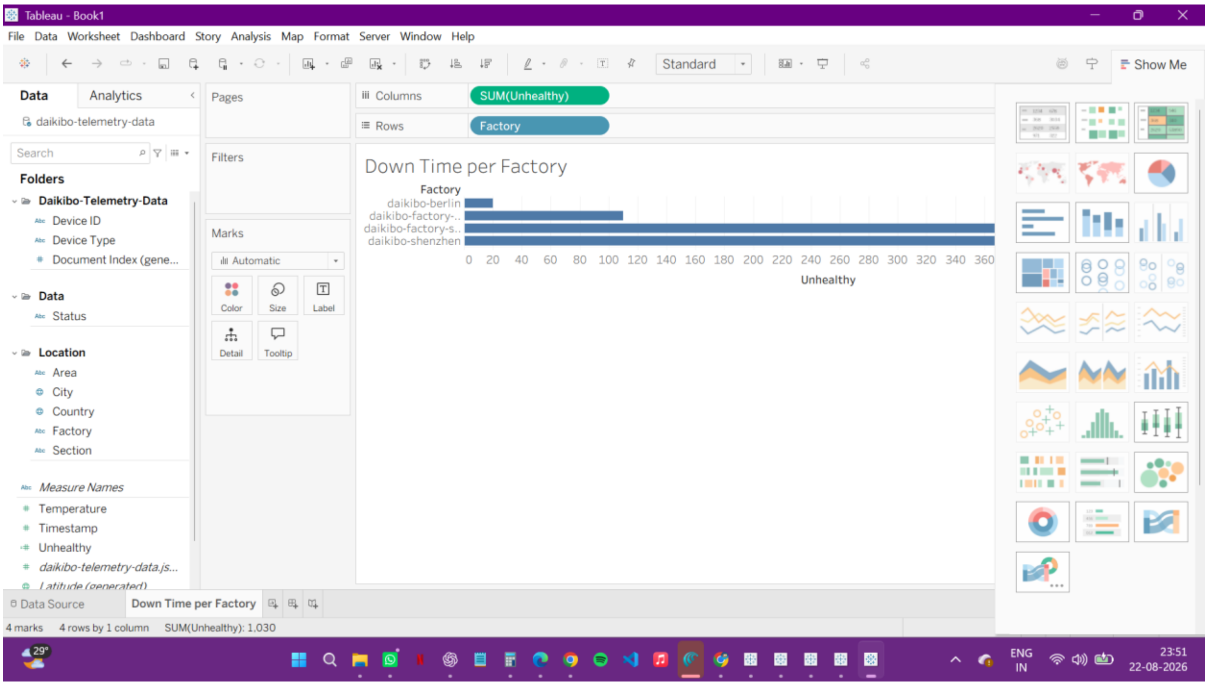
The dataset used in this task contained telemetry readings from Daikibo's factory equipment, including the health status recorded for each device at different points in time. Each record indicated whether a device was reporting a healthy or unhealthy status. I downloaded and imported this dataset into Tableau as the first step of the task.

Creation of the "Unhealthy" Calculated Measure

To quantify downtime from the raw health status readings, I created a calculated measure in Tableau named "Unhealthy". As instructed in the task, this calculated field assigned a value of 10 to every record where the device status was recorded as unhealthy, representing 10 minutes of potential downtime since the previous status message. Records with a healthy status contributed no downtime value. This calculated field formed the basis for both bar charts created later in the task.

Down Time per Factory

Using the "Unhealthy" calculated measure, I created a bar chart titled "Down Time per Factory". This chart aggregated the total downtime value across all devices within each factory, allowing a direct visual comparison of how much potential downtime each factory had accumulated.



Down Time per Device Type

I then created a second bar chart, titled "Down Time per Device Type", which aggregated the same downtime measure across different categories of devices rather than by factory. This chart made it possible to see which types of equipment were contributing most to overall downtime.

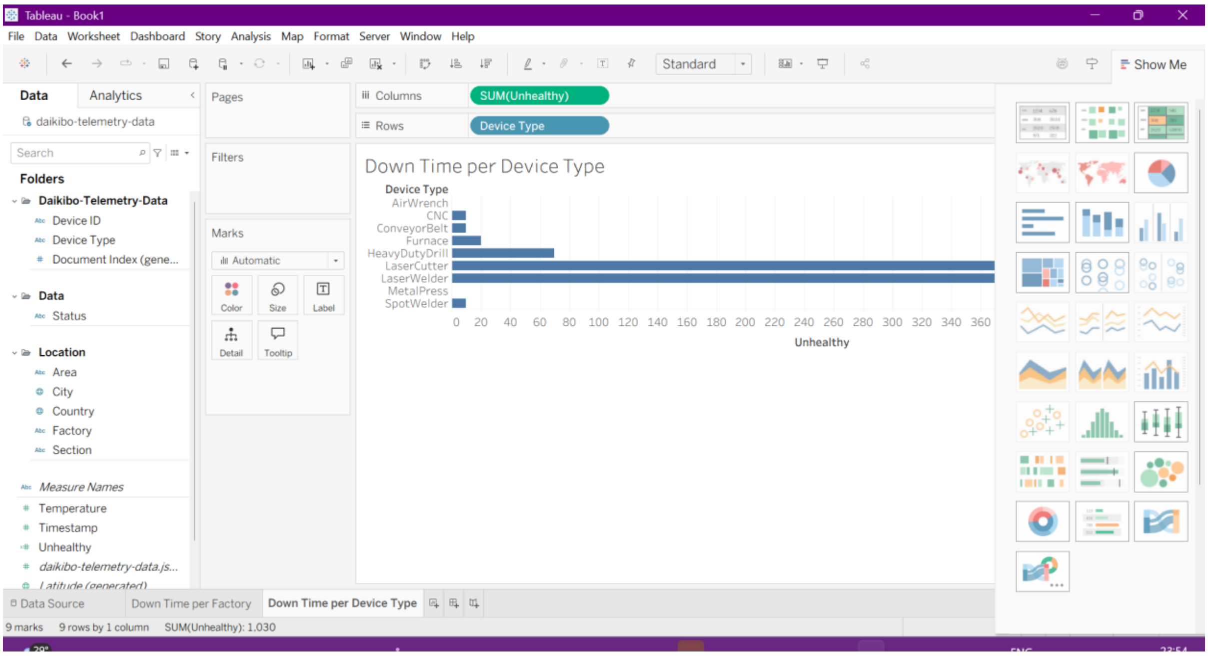
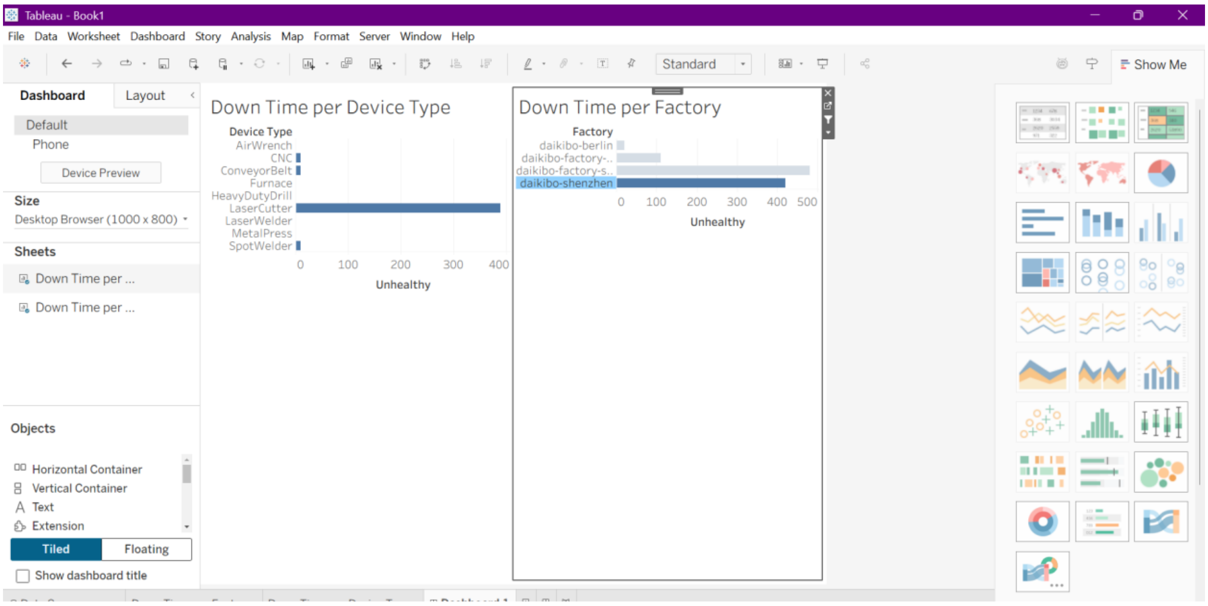
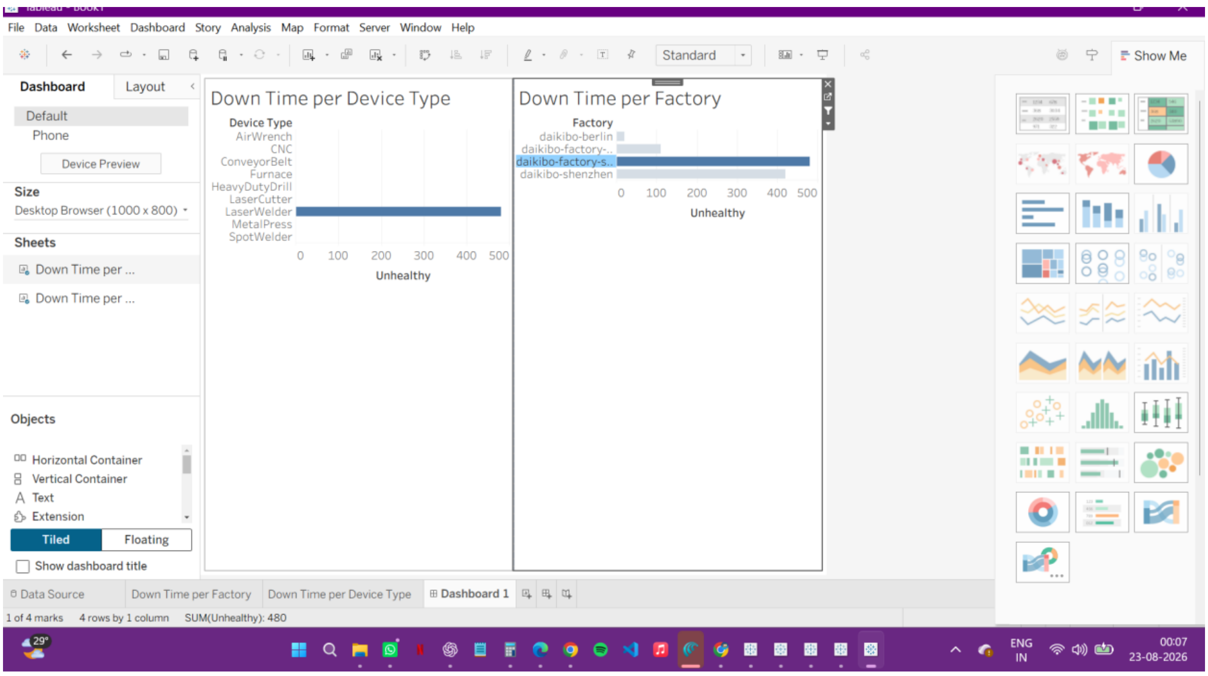


Tableau Dashboard and Filter Interaction

Once both charts were completed, I combined them into a single dashboard, named "Dashboard 1". As required by the task, I configured the "Down Time per Factory" chart to act as a filter for the "Down Time per Device Type" chart. This meant that selecting a specific factory in the first chart would automatically update the second chart to display only the device-type downtime relevant to that selected factory.



After building the filter interaction, I selected the factory with the highest overall downtime on the dashboard and captured a screenshot of the resulting filtered view, as required by the task submission instructions.



Task 1 Findings/Observations

Working through this task helped me understand how a simple calculated field can convert raw status data into a quantifiable business metric such as downtime. I observed that presenting the same underlying data at two different levels of granularity — by factory and by device type — made it possible to identify not only where downtime was occurring, but also what kind of equipment was most associated with it.

I also found that configuring one chart as a filter for another was an effective way to let a dashboard viewer drill down from a high-level comparison into more specific, factory-level detail, without needing to build a separate view for every factory.

Task 2 – Forensic Technology

The second task required me to work with an Excel dataset named "Equality Table.xlsx" as part of a forensic technology exercise. The purpose of this task was to classify equality scores recorded for different job roles across factories, in order to flag potential instances of unfair or discriminatory patterns in the underlying data.

Equality Score Dataset

The dataset contained three columns: Factory, Job Role, and Equality Score. The Equality Score for each row represented a numerical value indicating the degree of deviation from a neutral or fair baseline, with positive and negative values representing deviations in opposite directions.

Equality Class Classification

The task required me to create a fourth column, "Equality class", to categorise each row based on its Equality Score. The classification boundaries specified in the task were as follows:

Equality Class	**Condition (based on absolute value of score)**
Fair	Score from -10 to +10 (inclusive)
Unfair	Score below -10 or above +10, up to and including -20/+20
Highly Discriminative	Score below -20 or above +20

Excel Formula and Implementation

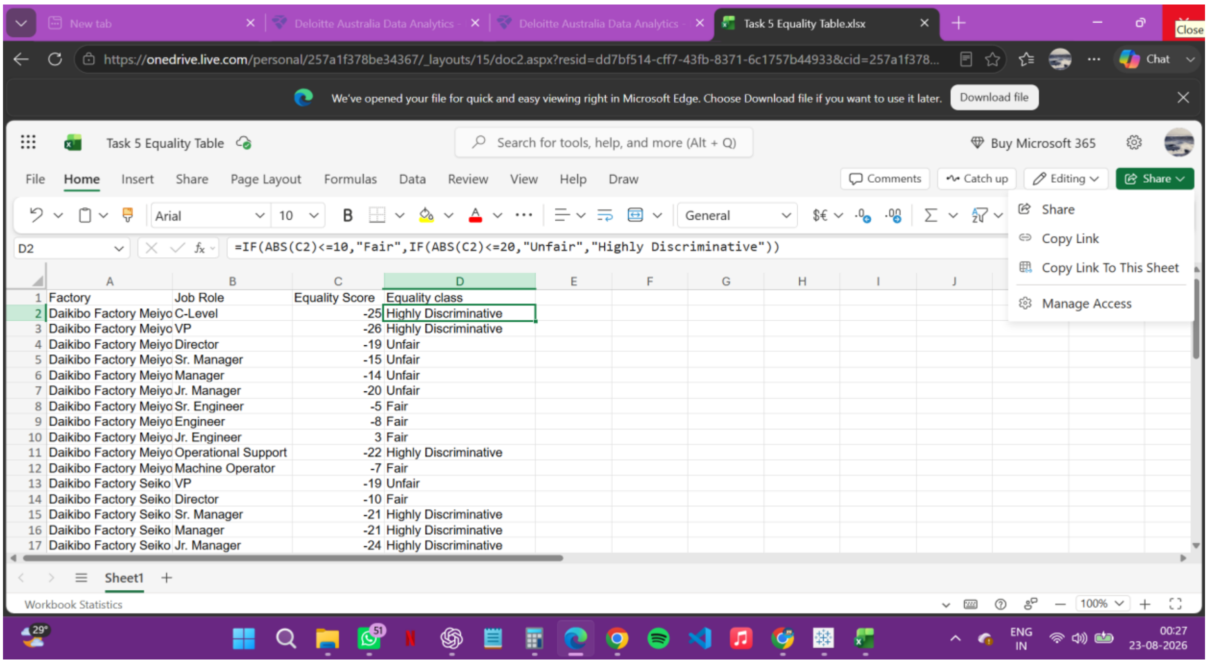
To implement this classification consistently across the dataset, I used the following nested IF formula in Excel, applied to the Equality Score column (Column C):

```
=IF(ABS(C2)<=10,"Fair",IF(ABS(C2)<=20,"Unfair","Highly
Discriminative"))
```

This formula first takes the absolute value of the Equality Score, so that positive and negative deviations of the same magnitude are treated identically. It then checks whether this absolute value falls within 10, in which case the row is classified as "Fair". If not, it checks whether the absolute value falls within 20, classifying the row as "Unfair". Any remaining rows, where the absolute value exceeds 20, are classified as "Highly Discriminative".

Applying this formula to sample scores illustrates the classification logic:

Equality Score	Absolute Value	Equality Class
10	10	Fair
-9	9	Fair
-19	19	Unfair
-30	30	Highly Discriminative



Task 2 Findings/Observations

This task demonstrated how a relatively simple conditional formula can be used to systematically flag potentially discriminatory patterns across a large dataset, without requiring manual review of every individual row. I observed that using the absolute value of the score was essential to the logic of the formula, since it ensured that large deviations were flagged consistently regardless of whether they were positive or negative.

I also noted the importance of applying classification boundaries exactly as specified in the task instructions, since even small differences in how a boundary condition is written (for example, using a strict inequality instead of "less than or equal to") could change the classification of borderline scores such as 10 or -20.

Skills Developed

Completing this simulation helped me develop the following skills:

- Building calculated fields and bar chart visualisations in Tableau.
- Designing an interactive dashboard with filter-based interactivity between multiple charts.
- Writing and applying nested conditional (IF) formulas in Microsoft Excel to classify data.
- Interpreting classification boundaries carefully and applying them consistently across a dataset.
- Structuring and documenting analytical work for submission through a GitHub repository.

Learning Outcomes

Beyond the specific technical skills listed above, this simulation gave me a clearer understanding of how data analytics is applied in a professional consulting context. I learned that a large part of data analytics work involves translating a business question — such as "which factory has the most downtime?" or "which scores indicate unfair treatment?" — into a well-defined calculation or formula that can be applied consistently across a dataset.

I also learned the value of building visualisations that are not just accurate but also easy for a non-technical audience to interpret, such as using an interactive dashboard filter instead of requiring a viewer to manually cross-reference two separate charts.

Challenges and Problem-Solving

One of the challenges I encountered while working on Task 1 was ensuring that the "Unhealthy" calculated field correctly reflected the intended downtime logic before building the charts on top of it, since any error in the calculated field would have affected both bar charts and the dashboard. I addressed this by checking the calculated field against the underlying data before proceeding to chart creation.

In Task 2, the main challenge was interpreting the classification boundaries precisely, particularly at the edges of each range (for example, a score of exactly 10 or exactly 20). I resolved this by working through the nested IF formula step by step and testing it against sample values to confirm that borderline scores were classified as intended before applying it to the full dataset.

Conclusion

The Deloitte Australia Data Analytics Job Simulation, completed through Forage, provided a practical introduction to the type of analytical work carried out by data analysts in a consulting environment. Through the two tasks in this simulation, I gained hands-on experience with Tableau for data visualisation and dashboard design, and with Microsoft Excel for conditional data classification.

While this was a virtual simulation rather than direct employment with Deloitte, it gave me a realistic sense of how analytical tools are applied to real business questions, and it strengthened my confidence in working independently through structured, instruction-based analytical tasks. I intend to continue building on the skills developed during this simulation as part of my ongoing academic and professional preparation in Computer Science and Engineering.

References

- Deloitte Australia Data Analytics Job Simulation, Forage. Available at: https://www.theforage.com/
- Deloitte. "About Deloitte." Available at: https://www2.deloitte.com/
- Certificate of Completion, Data Analytics Job Simulation, issued August 22, 2026 (Deloitte Australia via Forage).